

DS3231



DS3231 Real-Time Clock Module

Professional electronic component

v1.0

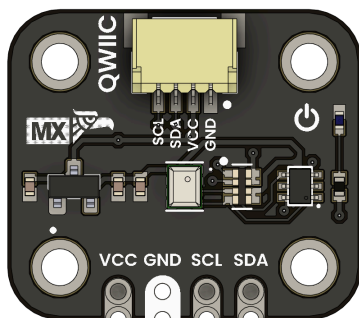
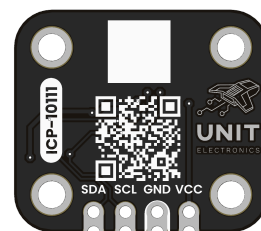
2025-09-09

Rev. A

PRODUCT OVERVIEW

DS3231 is a high-precision real-time clock (RTC) module that provides accurate timekeeping for embedded systems and microcontroller projects. It features a built-in temperature-compensated crystal oscillator (TCXO) to maintain precise time even in varying environmental conditions. The module communicates via the I2C interface, making it easy to integrate into a wide range of applications.

PRODUCT VIEWS

TOP VIEW*Component placement and connectors***BOTTOM VIEW***Underside components and connections*

KEY TECHNICAL SPECIFICATIONS



CONNECTIVITY

Primary Interface:**GPIO (Interrupt)**

Connector Type:**JST 4-pin 1.0mm**

Logic Levels:**VCC-referenced (2V – 5.5V tolerant)**

PIN CONFIGURATION

FUNCTION	NOTES
Power Supply	3.3V or 5V
Ground	Common ground for all components

KEY FEATURES



Digital Communication

I²C and SPI interfaces for easy integration



AI-Enabled Gas Classification

Support for Bosch BME AI-Studio for custom gas analysis

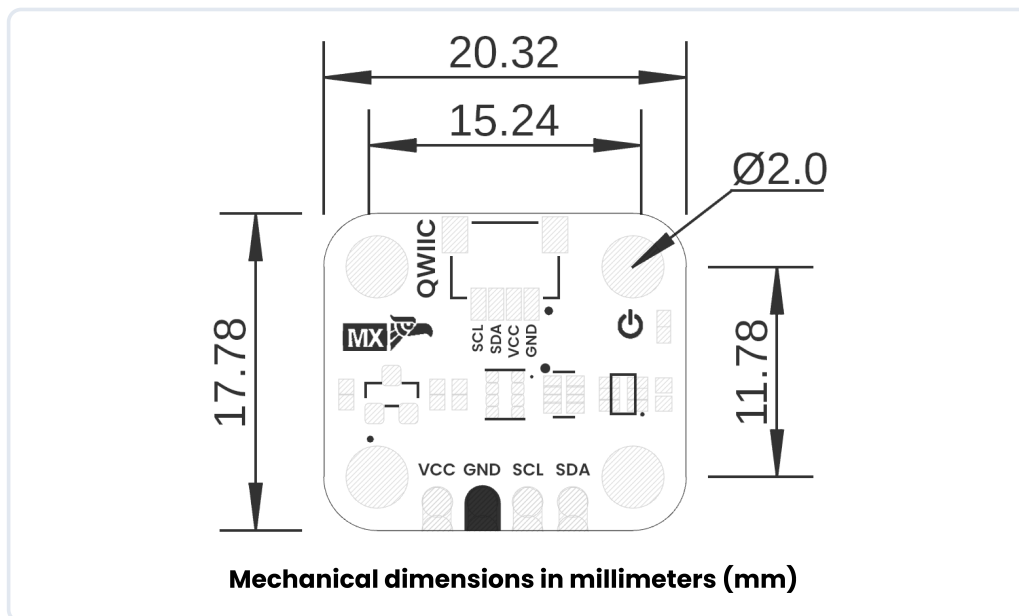
ADDITIONAL TECHNICAL INFORMATION

TYPICAL APPLICATIONS

APPLICATION	DESCRIPTION
Embedded Systems	Provides accurate timekeeping for microcontroller projects.
Data Logging	Timestamp data entries for logging applications.
IoT Devices	Maintain accurate time for scheduling and events.
Wearable Technology	Keep track of time in wearable devices.
Consumer Electronics	Used in devices like cameras and smart home systems.

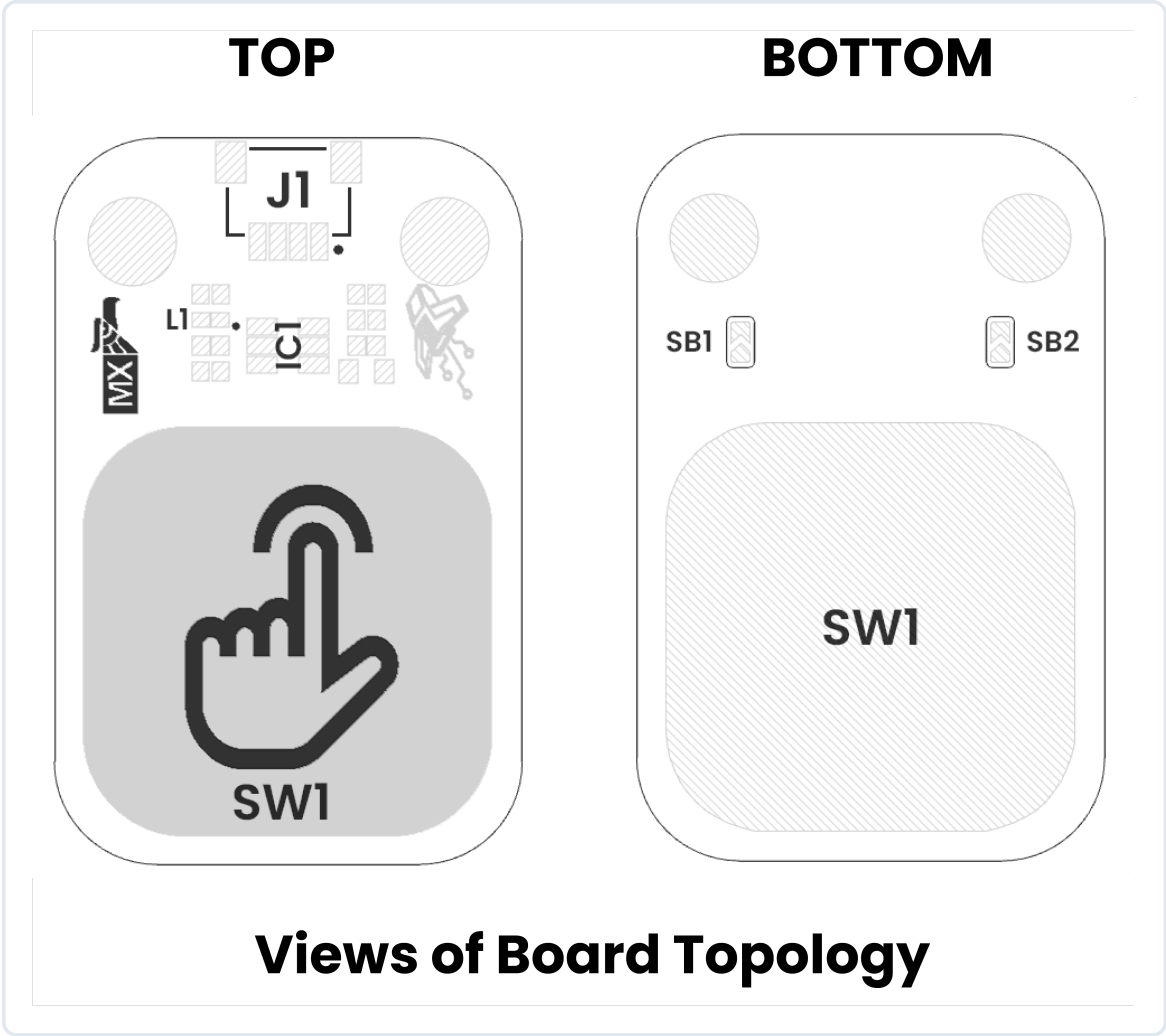
HARDWARE DOCUMENTATION

MECHANICAL DIMENSIONS



Physical dimensions and mounting specifications (measurements in millimeters)

SYSTEM TOPOLOGY



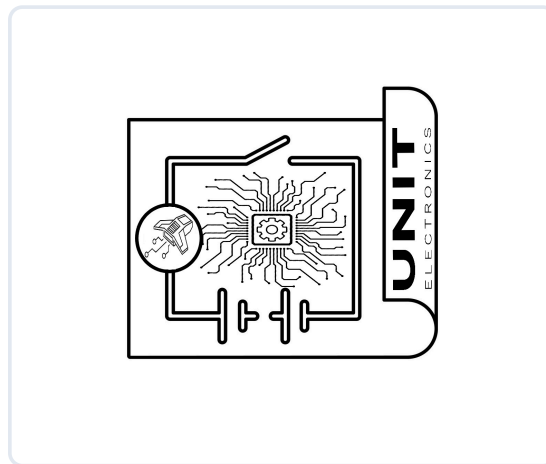
Connection topology and system integration diagram

Click image to open in full size

COMPONENT REFERENCE

REF.	DESCRIPTION
IC1	{{sensor_description}}
L1	Power On LED
U1	{{regulator_description}}
JP1	2.54 mm Castellated Holes
J1	QWIIC Connector (JST 1 mm pitch) for I2C

CIRCUIT SCHEMATIC



Complete circuit schematic showing all component connections

PIN DESCRIPTION

Detailed pin assignment and electrical specifications

SIGNAL DESCRIPTION

FUNCTION	NOTES
Power Supply	3.3V or 5V
Ground	Common ground for all components

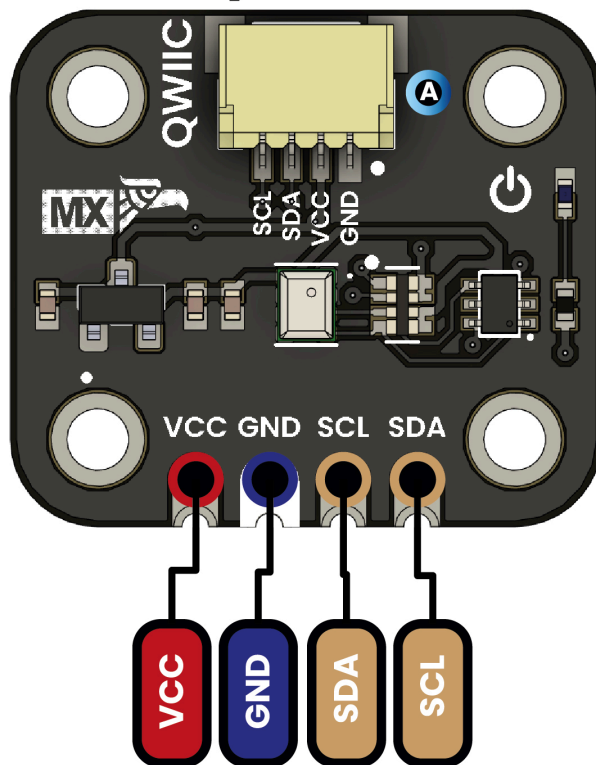
VOLTAGE LEVEL	FUNCTION
3.3 V – 5.5 V	Provides power to the on-board regulator and sensor core.
0 V	Common reference for power and signals.
1.8 V to VCC	Serial data line for I ² C communications.
1.8 V to VCC	Serial clock line for I ² C communications.

PIN CONFIGURATION LAYOUT

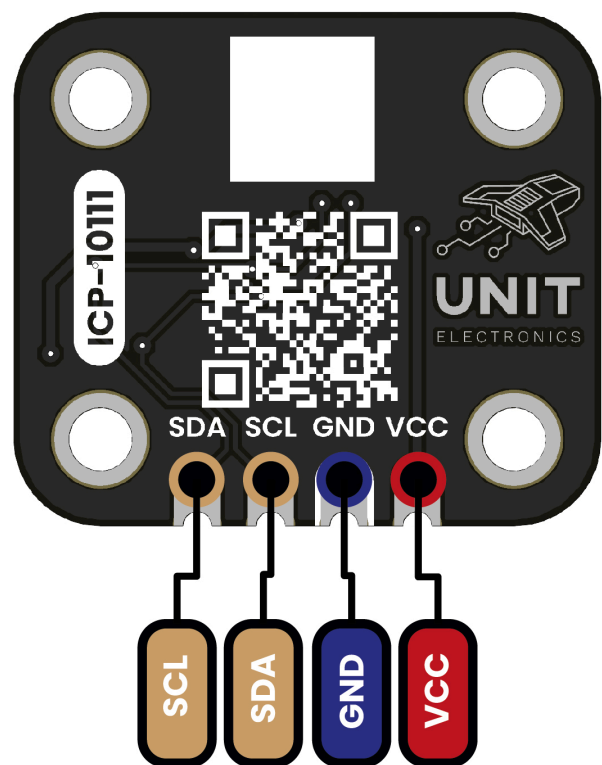
Physical connector layout and pin positioning

PINOUT

Top view



Bottom view



Description:

 Supply voltage

 GND

 I2C

 **QWIIIC**®

Complete pin configuration diagram showing all connectors, pin assignments, and electrical connections for proper integration